

System Assumptions & Constraints

Team 01

Assumptions: Environment

- Density, $\rho = 1.225 \text{ kg/m}^3$
 - Density at sea level, roughly where entire mission takes place
 - From Anderson, standard atmosphere tables
- Gravitational acceleration, $g = 9.81 \text{ m/s}^2$
 - Known constant gravitational acceleration value
 - NIST Fundamental Physical Constants
- Takeoff friction coefficient, $\mu_{\text{TO}} = 0.05$
 - From Gudmundsson, table 18-4
 - Value used for concrete runway, no brakes
- Landing friction coefficient, $\mu_{\text{landing}} = 0.08$
 - From Gudmundsson, table 18-4
 - Value used for wet grass, no brakes

Assumptions: Performance

- $\phi_{\text{cruise}} = 0.75$
 - Sadraey, chapter 4 page 126
- $\gamma = 16.7$ degrees
 - Calculated from mission altitude and estimated distance for climb
 - Mission altitude: 30m
 - Estimated climb distance: 100m
- Span efficiency, $e = 0.7$
 - Selected conservative value from range given by Sadraey, chapter 4 page 122
- $AR = 5$
 - Selected aspect ratio of wing based on aircraft type and stall speed
 - From Sadraey, table 5.8

$$\gamma = \tan^{-1} \left(\frac{h_{\text{cruise}}}{L_{\text{climb}}} \right)$$

Assumptions: Performance

- Max lift coefficient, $C_{L,max} = 1.5$
 - Selected from Sadraey, table 4.11 for home-built aircraft
- Cruise lift coefficient, $C_{L,C} = 0.3$
 - Typical value for subsonic aircraft, Sadraey, chapter 4 page 132
- Takeoff lift coefficient, $C_{L,TO} = 0.3$
 - Before rotation and no flaps, therefore similar to cruise
- Lift coefficient at rotation, $C_{LR} = 1.24$
 - Calculated from $C_{L,max}$ and takeoff speed, 1.1 from takeoff velocity multiplier, Sadraey
- $C_{D,0} = 0.04$
 - Selected from Sadraey, table 4.12 for home-built/GA with fixed gear
- $C_D = 0.048$
 - Calculated from $C_{D,0}$, $C_{L,C}$, and induced drag factor
 - Sadraey, equation 4.41
 - $K = 0.091$ induced drag factor, from Sadraey, eqn 4.41

$$C_{LR} = \frac{C_{L,max}}{1.1^2}$$

$$C_D = C_{D0} + KC_{LC}^2$$

Assumptions: Performance

- Takeoff drag coefficient, $C_{D,TO} = 0.048$
 - Before rotation and no flaps, therefore similar to cruise
- Ground roll drag coefficient, $C_{DG} = 0.033$
 - Sadraey, equation 4.67a
- Load factor, $n = 1.5$
 - Selected conservative value from recommended range for RC planes
 - From Sadraey, table 10.9
- $\eta_p = 0.75$
 - From Gudmundsson, table 3-3 for fixed pitch propellor
- $\eta_{p,TO} = 0.45$
 - From Gudmundsson, table 3-3 for fixed pitch propellor, average efficiency during takeoff

Assumptions: Performance

- Stall speed, $V_{\text{stall}} = 10.5 \text{ m/s}$
 - Selected middle value from recommended stall speed range
 - Range from A2 Handout from class
- Cruise speed, $V_{\text{cruise}} = 23.5 \text{ m/s}$
 - Calculated from $C_{L,\text{max}}$, $C_{L,C}$, and stall speed
- Takeoff speed, $V_{\text{TO}} = 11.6 \text{ m/s}$
 - From Sadraey, chapter 4, page 132
 - Used $1.1 * V_{\text{stall}}$ as conservative value for takeoff velocity
- $R_{\text{turn}} = 20 \text{ m}$
 - Selected value smaller than required turn radius for mission flight path that also allowed for turn speed between stall and cruise speeds
- Maneuver speed, $V_M = 14.8 \text{ m/s}$
 - Calculated from selected turn radius and load factor

$$\frac{1}{2}\rho V_s^2 C_{L_{\text{max}}} S = \frac{1}{2}\rho V_{\text{cruise}}^2 C_{L_{\text{cruise}}} S$$

Cancel common factors ($\frac{1}{2}$, ρ , and S):

$$V_s^2 C_{L_{\text{max}}} = V_{\text{cruise}}^2 C_{L_{\text{cruise}}}$$

Solve for cruise velocity (V_{cruise}):

$$V_{\text{cruise}} = V_s \sqrt{\frac{C_{L_{\text{max}}}}{C_{L_{\text{cruise}}}}}$$

$$V_M = \sqrt{R_M g \sqrt{n^2 - 1}}$$

Constraints (From RFP)

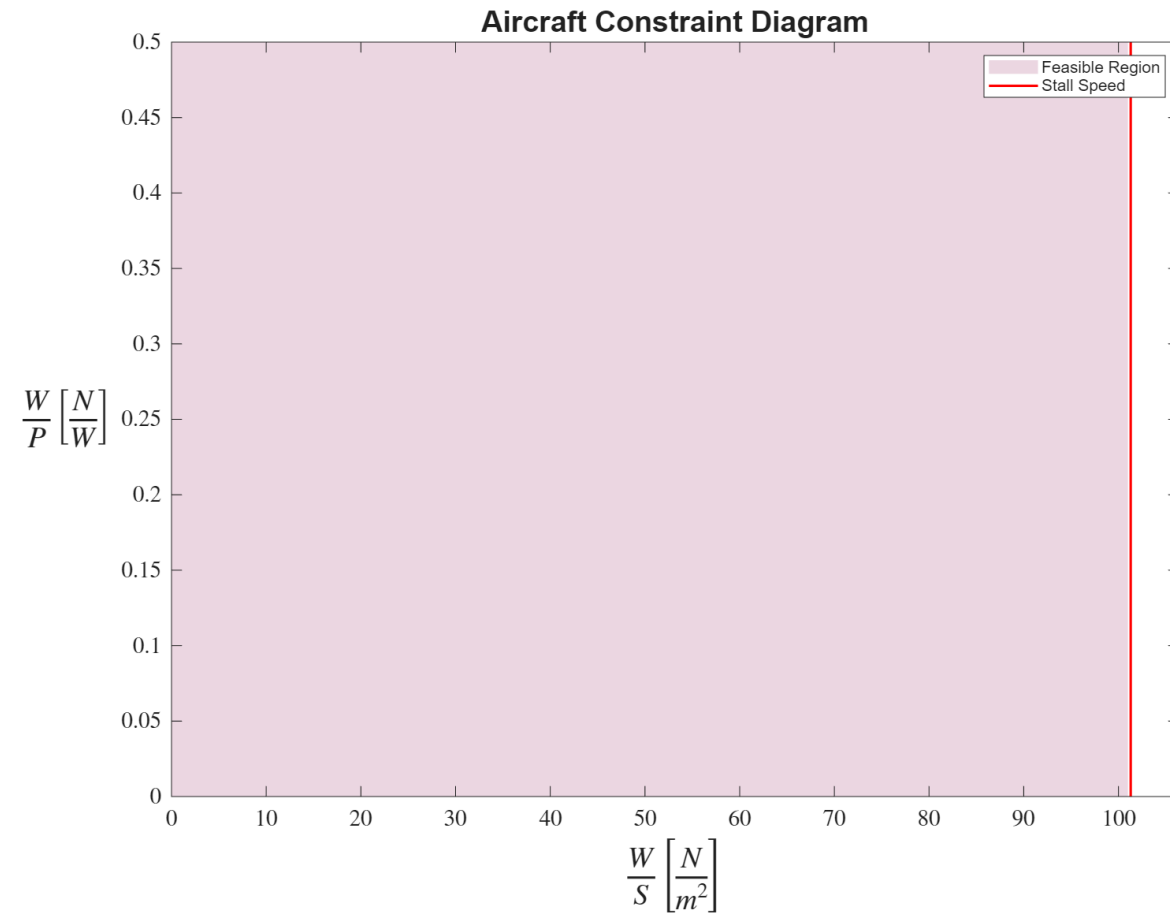
- Wingspan ≤ 1.5 m
- Aircraft Length ≤ 0.75 m
- Aircraft Height ≤ 0.75 m
- $M_{TO} \leq 6$ kg
- $h_{cruise} = 30$ m
- $R_{turn,max} = 50$ m
- $S_{TO} \leq 25$ m
- $P_{motor,max} \leq 1$ kW
- Stable with and without payload
- $M_{payload} = 520g \pm 10g$
- $L_{payload} = 50.8$ mm (± 1 mm)
- Cost $\leq \$400$



Constraints: Stall

- Sets upper bound for wing loading
- $\rho = 1.225 \text{ kg/m}^3$
- $C_{L,\max} = 1.5$
- $V_{\text{stall}} = 10.5 \text{ m/s}$
- W/S upper bound set to 101.29 N/m^2
- Enforces requirement(s): R5, R10, R22

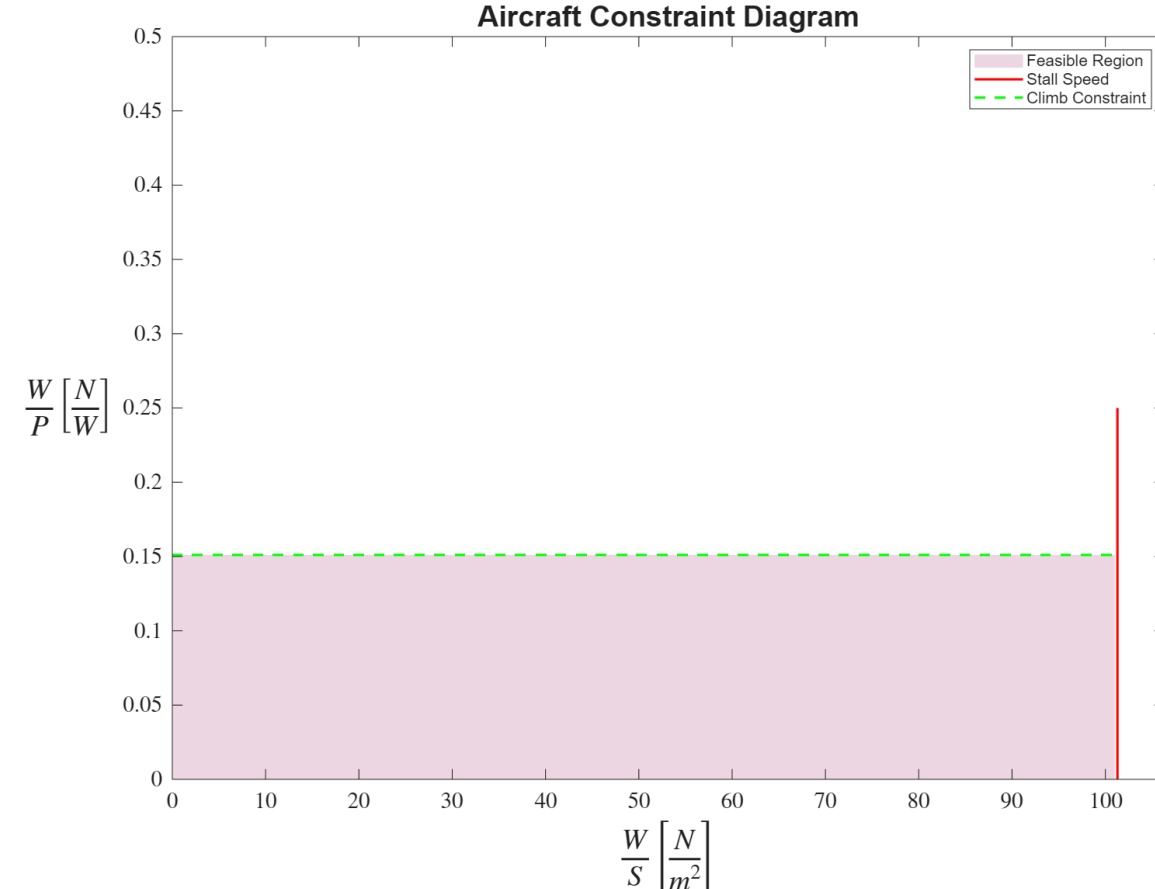
$$\frac{W}{S} = \frac{1}{2} \rho V_s^2 C_L^{\max}$$



Constraints: Climb

- Sets upper bound for power loading
- $\eta_p = 0.75$
- $V_{\text{stall}} = 10.5 \text{ m/s}$
- $\left(\frac{L}{D}\right)_{\text{max}} = 6.23$
- $\gamma = 16.7 \text{ degrees}$
- Power loading limit set to 0.15 N/W
- Enforces requirement(s): R2, R21, R22

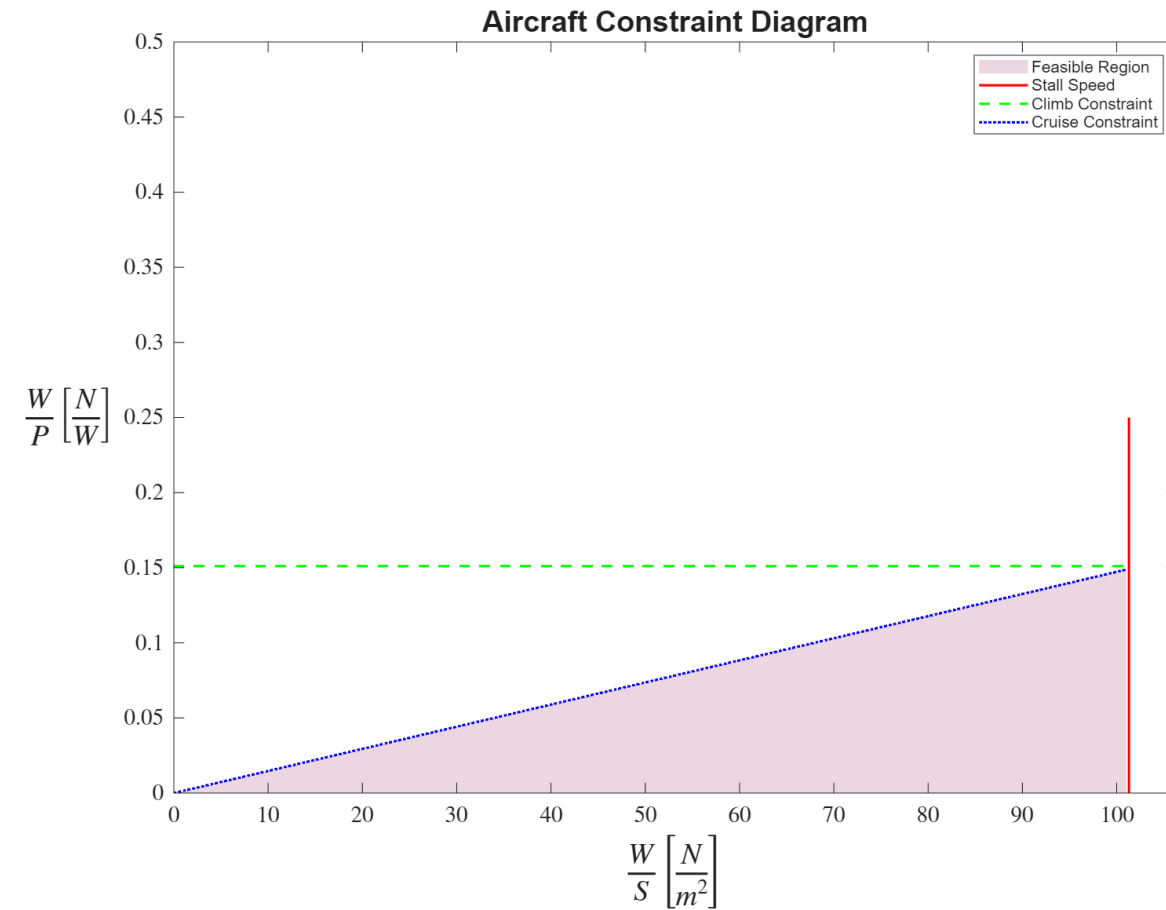
$$\frac{W}{P} = \frac{\eta_P}{V \left(\frac{1}{0.866(L/D)_{\text{max}}} + \sin \gamma \right)}$$



Constraints: Cruise

- Sets upper bound for power loading
- $\eta_p = 0.75$
- $V_{\text{cruise}} = 23.5 \text{ m/s}$
- $\phi_{\text{cruise}} = 0.75$
- $\rho = 1.225 \text{ kg/m}^3$
- $C_{D,0} = 0.04$
- Power loading limit set to 0.15 N/W at 101.29 N/m^2
- Enforces requirement(s): R3, R21

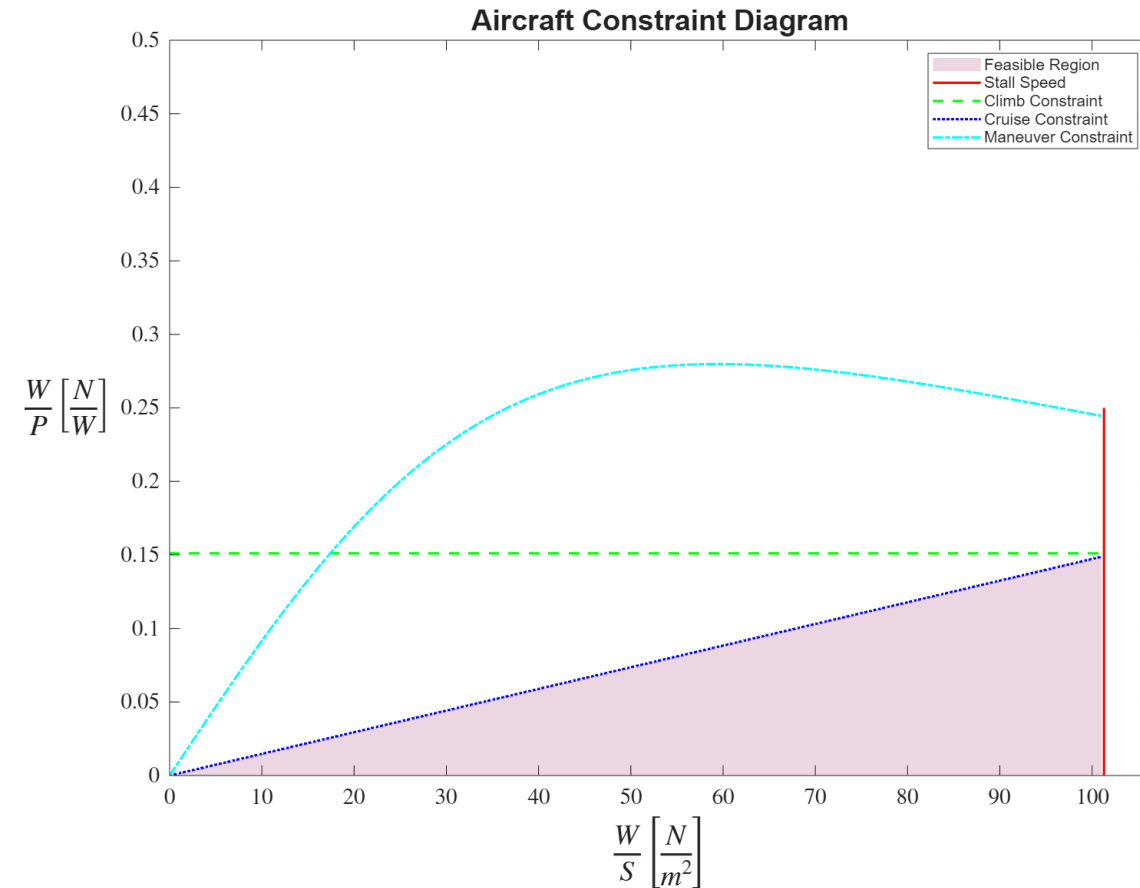
$$\frac{W}{P} = \left[\frac{\eta_P \cdot \Phi}{0.5(1.1)C_{D0} \cdot \rho \cdot V^3} \right] \frac{W}{S}$$



Constraints: Maneuver

- Sets upper bound for power loading in our case
- $\eta_p = 0.75$
- $q = 0.5 * \rho * V^2 = 134.3 \text{ Pa}$
- $C_{D,0} = 0.04$
- $V_M = 14.8 \text{ m/s}$
 - Minimum turn radius of 20 m allows for maneuverability off the runway
 - V_M found using $V_M = \sqrt{R_M g \sqrt{n^2 - 1}}$
 - V_M greater than $V_{M,\text{stall}} = 12.86$, so chosen turn speed is viable
 $V_{M,\text{stall}} = V_{\text{stall}} \sqrt{n}$
- $n = 1.5$
- $AR = 5$
- $e = 0.7$
- Does not impact feasible region
- Enforces requirement(s): R3, R4, R19

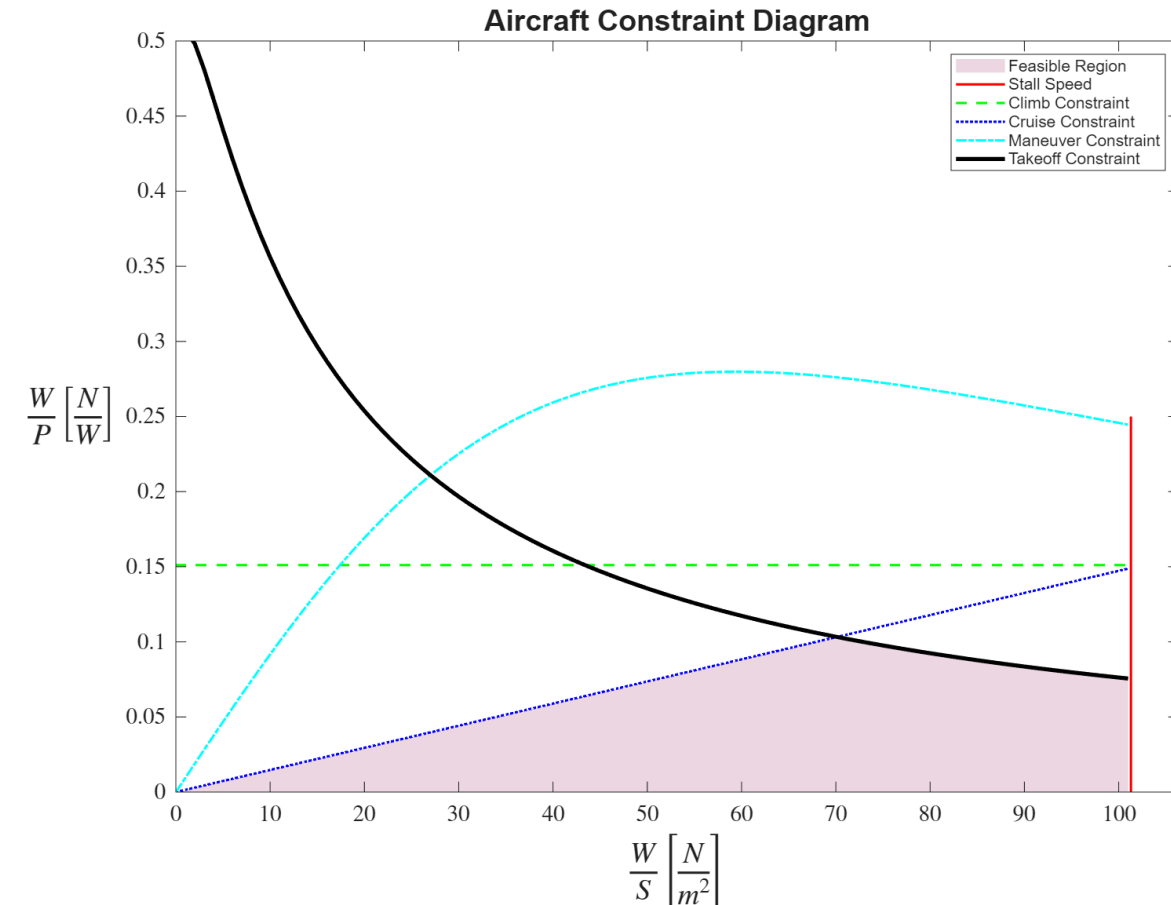
$$\frac{W}{P} = \frac{\eta_P}{qV \left[\frac{C_{D0}}{W/S} + \frac{n^2}{q^2 \pi A Re} \frac{W}{S} \right]}$$



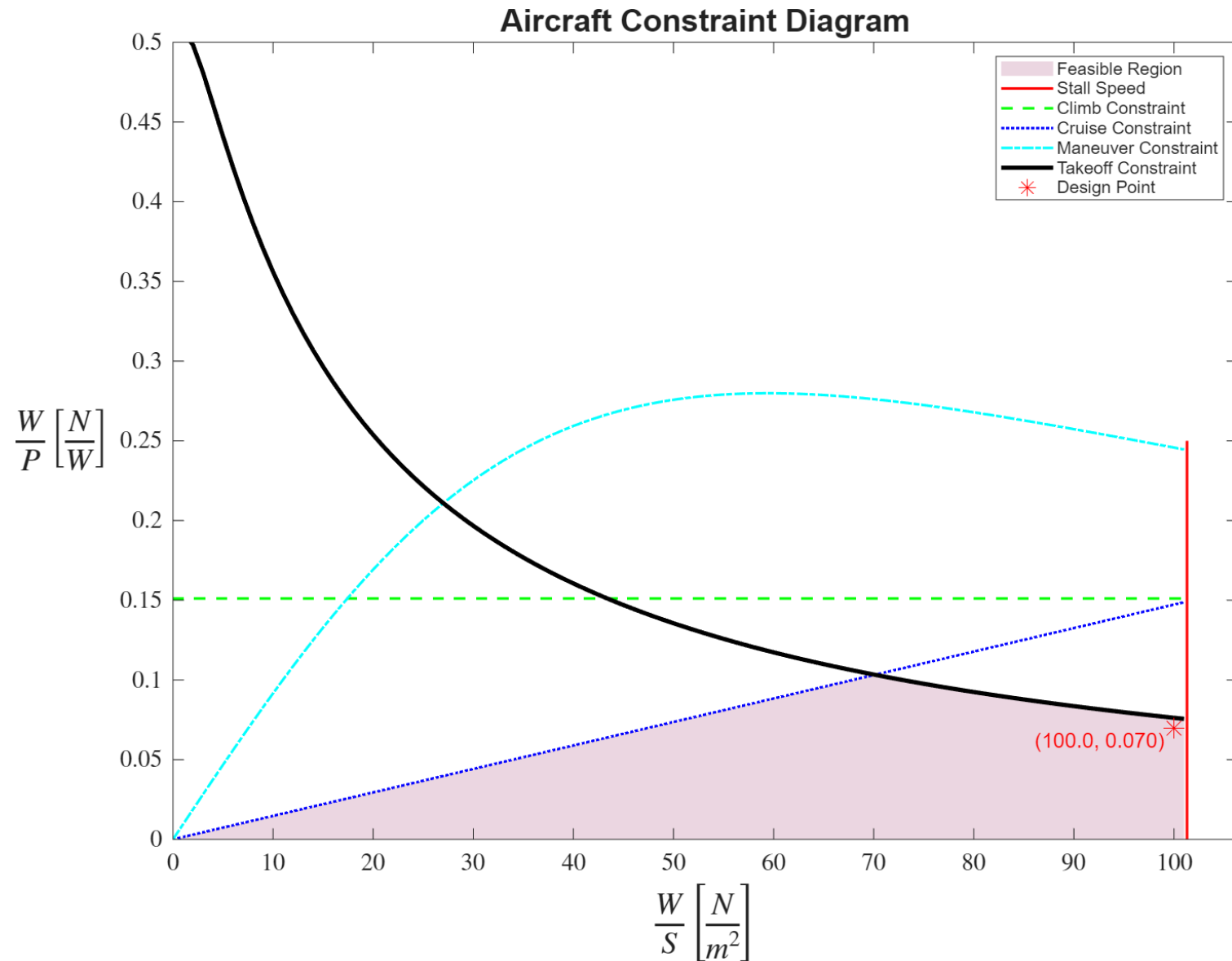
Constraints: Take-Off

- Sets upper bound for power loading
- $\eta_{p,TO} = 0.45$
- $\rho = 1.225 \text{ kg/m}^3$
- $g = 9.81 \text{ m/s}^2$
- $C_{DG} = 0.033$
- $C_{LR} = 1.24$
- $\mu_{TO} = 0.05$
- $S_{TO} = 25 \text{ m}^2$
- $V_{TO} = 11.6 \text{ m/s}$
- Power loading limit set to $\sim 0.07 \text{ N/W}$ at 101.29 N/m^2
- Enforces Requirement(s): R1, R21, R22

$$\left(\frac{W}{P}\right)_{STO} = \frac{1 - \exp\left(0.6\rho g C_{DG} S_{to} \frac{1}{W/S}\right)}{\mu - \left(\mu - \frac{C_{DG}}{C_{LR}}\right) \left[\exp\left(0.6\rho g C_{DG} S_{to} \frac{1}{W/S}\right)\right]} \frac{\eta_P}{V_{to}}$$



Design Point



Design Point

- Wing loading selected at 100 N/m^2
 - Near maximum for assumed stall speed
 - Minimizes wing area in feasible design space, reduces cost
- Power loading selected at 0.07 N/W
 - Near maximum in design space for selected wing loading
 - Minimizes power requirement
- Chose design point near takeoff and stall corner rather than near takeoff and climb corner because this allows for much smaller wing area with relatively similar required power

Appendix

Code:

Code for this project can be found in our team's GitHub repo [here.](#)

Sources

Sadraey, M. H. (2012). *Aircraft Design: A Systems Engineering Approach*.

Gudmundsson, S. (2022). *General Aviation Aircraft Design*.

Anderson, J. D. (2016). *Introduction to Flight, Eighth Edition*.

AAE 451 Lecture Notes